

```
2026-08-22T09:59:20.161-04:00 INFO 28512 --- [banco-batch] [ main] c.duc.bancobatch.BancoBatchApplication : Starting BancoBatchApplication using Java 21.0.12.1 with PID 28512 (C:\Users\Personal\Desktop\Bac
kend III\Semana-2\EXPI_52_Grupo_1\banco-batch\Target\Classes started by Personal in C:\Users\Personal\Desktop\Backend III\Semana-2\EXPI_52_Grupo_1\banco-batch)
2026-08-22T09:59:20.166-04:00 INFO 28512 --- [banco-batch] [ main] c.duc.bancobatch.BancoBatchApplication : No active profile set, falling back to 1 default profile: "default"
2026-08-22T09:59:21.150-04:00 INFO 28512 --- [banco-batch] [ main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-08-22T09:59:21.624-04:00 INFO 28512 --- [banco-batch] [ main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Added connection org.postgresql.jdbc.PgConnection@5e8cdad75
2026-08-22T09:59:21.624-04:00 INFO 28512 --- [banco-batch] [ main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-08-22T09:59:22.038-04:00 INFO 28512 --- [banco-batch] [ main] c.duc.bancobatch.BancoBatchApplication : Started BancoBatchApplication in 2.464 seconds (process running for 2.929)
2026-08-22T09:59:22.044-04:00 INFO 28512 --- [banco-batch] [ main] o.s.b.b.a.JobLauncherApplicationRunner : Running default command line with: [fechaEjecucion=20260822095909]
2026-08-22T09:59:22.304-04:00 INFO 28512 --- [banco-batch] [ main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: {name=transaccionesJob}] launched with the following parameters: [[JobParameter{
name='fechaEjecucion', value=20260822095909, type=class java.lang.String, identifying=true}]]
2026-08-22T09:59:22.372-04:00 INFO 28512 --- [banco-batch] [ main] o.s.batch.core.step.AbstractStep : Executing step: [procesarTransaccionesStep]
2026-08-22T09:59:22.380-04:00 INFO 28512 --- [banco-batch] [ main] c.d.b.listener.BatchStepListener : STEP INICIADO | step=procesarTransaccionesStep | fechaInicio=2026-08-22T09:59:22.379689180
2026-08-22T09:59:22.406-04:00 WARN 28512 --- [banco-batch] [transacciones-3] c.d.b.listener.BatchSkipListener : REGISTRO OMITIDO | etapa=PROCESS | registro=TransaccionCsv[id=3, fecha='2024-01-03', monto=-200,
tipo='debito'] | motivo=Monto debe ser mayor que cero. ID: 3, monto: -200
2026-08-22T09:59:22.407-04:00 WARN 28512 --- [banco-batch] [transacciones-2] c.d.b.listener.BatchSkipListener : REGISTRO OMITIDO | etapa=PROCESS | registro=TransaccionCsv[id=4, fecha='2024-01-03', monto=0, tip
o='debito'] | motivo=Monto debe ser mayor que cero. ID: 4, monto: 0
2026-08-22T09:59:22.464-04:00 INFO 28512 --- [banco-batch] [ main] o.s.batch.core.step.AbstractStep : Step: [procesarTransaccionesStep] executed in 93ms
2026-08-22T09:59:22.478-04:00 INFO 28512 --- [banco-batch] [ main] c.d.b.listener.BatchStepListener : STEP FINALIZADO | step=procesarTransaccionesStep | status=COMPLETED | leidos=10 | escritos=8 | omitid
o=2
2026-08-22T09:59:22.529-04:00 INFO 28512 --- [banco-batch] [ main] o.s.batch.core.step.AbstractStep : Executing step: [generarResumenTransaccionesStep]
2026-08-22T09:59:22.530-04:00 INFO 28512 --- [banco-batch] [ main] c.d.b.listener.BatchStepListener : STEP INICIADO | step=generarResumenTransaccionesStep | fechaInicio=2026-08-22T09:59:22.530314100
2026-08-22T09:59:22.570-04:00 INFO 28512 --- [banco-batch] [ main] o.s.batch.core.step.AbstractStep : Step: [generarResumenTransaccionesStep] executed in 47ms
2026-08-22T09:59:22.582-04:00 INFO 28512 --- [banco-batch] [ main] c.d.b.listener.BatchStepListener : STEP FINALIZADO | step=generarResumenTransaccionesStep | status=COMPLETED | leidos=7 | escritos=7
2026-08-22T09:59:22.582-04:00 INFO 28512 --- [banco-batch] [ main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: {name=transaccionesJob}] completed with the following parameters: [[JobParameter{
name='fechaEjecucion', value=20260822095909, type=class java.lang.String, identifying=true}]] and the following status: [COMPLETED] in 295ms
|
```

La captura registra las métricas de ejecución del procesamiento de transacciones con multithreading, permitiendo comprobar el volumen procesado y el resultado del Step.

```
2026-08-22T10:13:15.237-04:00 INFO 20240 --- [banco-batch] [ main] c.duc.bancobatch.BancoBatchApplication : Starting BancoBatchApplication using Java 21.0.12.1 with PID 20240 (C:\Users\Personal\Desktop\Bac
kend III\Semana-2\EXPI_52_Grupo_1\banco-batch\Target\Classes started by Personal in C:\Users\Personal\Desktop\Backend III\Semana-2\EXPI_52_Grupo_1\banco-batch)
2026-08-22T10:13:15.243-04:00 INFO 20240 --- [banco-batch] [ main] c.duc.bancobatch.BancoBatchApplication : No active profile set, falling back to 1 default profile: "default"
2026-08-22T10:13:16.272-04:00 INFO 20240 --- [banco-batch] [ main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-08-22T10:13:16.695-04:00 INFO 20240 --- [banco-batch] [ main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection org.postgresql.jdbc.PgConnection@3e1a3801
2026-08-22T10:13:16.698-04:00 INFO 20240 --- [banco-batch] [ main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-08-22T10:13:17.018-04:00 INFO 20240 --- [banco-batch] [ main] c.duc.bancobatch.BancoBatchApplication : Started BancoBatchApplication in 2.487 seconds (process running for 2.987)
2026-08-22T10:13:17.023-04:00 INFO 20240 --- [banco-batch] [ main] o.s.b.b.a.JobLauncherApplicationRunner : Running default command line with: [fechaEjecucion=20260822101305]
2026-08-22T10:13:17.023-04:00 INFO 20240 --- [banco-batch] [ main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: {name=interesesJob}] launched with the following parameters: [[JobParameter{
name='fechaEjecucion', value=20260822101305, type=class java.lang.String, identifying=true}]]
2026-08-22T10:13:17.283-04:00 INFO 20240 --- [banco-batch] [ main] o.s.batch.core.step.AbstractStep : Executing step: [procesarInteresesStep]
2026-08-22T10:13:17.285-04:00 INFO 20240 --- [banco-batch] [ main] c.d.b.listener.BatchStepListener : STEP INICIADO | step=procesarInteresesStep | fechaInicio=2026-08-22T10:13:17.285911900
2026-08-22T10:13:17.323-04:00 WARN 20240 --- [banco-batch] [ intereses-3] c.d.b.listener.BatchSkipListener : REGISTRO OMITIDO | etapa=PROCESS | registro=InteresCsv[cuentaId=105, nombre='Charlie Green', sald
o=500, edad=35, tipo='hipoteca'] | motivo=Tipo de cuenta no permitido. hipotec
2026-08-22T10:13:17.323-04:00 WARN 20240 --- [banco-batch] [ intereses-2] c.d.b.listener.BatchSkipListener : REGISTRO OMITIDO | etapa=PROCESS | registro=InteresCsv[cuentaId=104, nombre='Alice Brown', sald
o=500, edad=45, tipo='ahorro'] | motivo=Saldo debe ser mayor que cero. Cuenta ID: 104, saldo: 0
2026-08-22T10:13:17.380-04:00 INFO 20240 --- [banco-batch] [ main] o.s.batch.core.step.AbstractStep : Step: [procesarInteresesStep] excuted in 105ms
2026-08-22T10:13:17.394-04:00 INFO 20240 --- [banco-batch] [ main] c.d.b.listener.BatchStepListener : STEP FINALIZADO | step=procesarInteresesStep | status=COMPLETED | leidos=8 | escritos=6 | omitid
o=2
2026-08-22T10:13:17.437-04:00 INFO 20240 --- [banco-batch] [ main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: {name=interesesJob}] completed with the following parameters: [[JobParameter{
name='fechaEjecucion', value=20260822101305, type=class java.lang.String, identifying=true}]] and the following status: [COMPLETED] in 108ms
|
```

La captura presenta las métricas obtenidas durante el procesamiento paralelo del Job de intereses y evidencia la finalización de su ejecución.

```
2026-08-22T10:18:12.381-04:00 INFO 7596 --- [banco-batch] [ main] c.duc.bancobatch.BancoBatchApplication : Starting BancoBatchApplication using Java 21.0.12.1 with PID 7596 (C:\Users\Personal\Desktop\Bac
kend III\Semana-2\EXPI_52_Grupo_1\banco-batch\Target\Classes started by Personal in C:\Users\Personal\Desktop\Backend III\Semana-2\EXPI_52_Grupo_1\banco-batch)
2026-08-22T10:18:12.388-04:00 INFO 7596 --- [banco-batch] [ main] c.duc.bancobatch.BancoBatchApplication : No active profile set, falling back to 1 default profile: "default"
2026-08-22T10:18:13.432-04:00 INFO 7596 --- [banco-batch] [ main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-08-22T10:18:13.847-04:00 INFO 7596 --- [banco-batch] [ main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection org.postgresql.jdbc.PgConnection@42c28305
2026-08-22T10:18:13.850-04:00 INFO 7596 --- [banco-batch] [ main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-08-22T10:18:14.336-04:00 INFO 7596 --- [banco-batch] [ main] c.duc.bancobatch.BancoBatchApplication : Started BancoBatchApplication in 2.64 seconds (process running for 3.164)
2026-08-22T10:18:14.341-04:00 INFO 7596 --- [banco-batch] [ main] o.s.b.b.a.JobLauncherApplicationRunner : Running default command line with: [fechaEjecucion=20260822101803]
2026-08-22T10:18:14.539-04:00 INFO 7596 --- [banco-batch] [ main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: {name=estadosAnualesJob}] launched with the following parameters: [[JobParameter{
name='fechaEjecucion', value=20260822101803, type=class java.lang.String, identifying=true}]]
2026-08-22T10:18:14.602-04:00 INFO 7596 --- [banco-batch] [ main] o.s.batch.core.step.AbstractStep : Executing step: [procesarMovimientosAnualesStep]
2026-08-22T10:18:14.605-04:00 INFO 7596 --- [banco-batch] [ main] c.d.b.listener.BatchStepListener : STEP INICIADO | step=procesarMovimientosAnualesStep | fechaInicio=2026-08-22T10:18:14.605183300
2026-08-22T10:18:14.694-04:00 INFO 7596 --- [banco-batch] [ estados anuales-2] c.d.b.listener.BatchSkipListener : REGISTRO OMITIDO | etapa=PROCESS | registro=MovimientoAnual[cuentaId=107, fecha='2024-12-29', t
ransaccion=deposito, monto=0, descripcion='Ingreso navideño'] | motivo=Monto no puede ser cero. Cuenta ID: 107
2026-08-22T10:18:14.702-04:00 INFO 7596 --- [banco-batch] [ main] o.s.batch.core.step.AbstractStep : Step: [procesarMovimientosAnualesStep] executed in 108ms
2026-08-22T10:18:14.716-04:00 INFO 7596 --- [banco-batch] [ main] c.d.b.listener.BatchStepListener : STEP FINALIZADO | step=procesarMovimientosAnualesStep | status=COMPLETED | leidos=9 | escritos=8 | omitid
o=1
2026-08-22T10:18:14.770-04:00 INFO 7596 --- [banco-batch] [ main] o.s.batch.core.step.AbstractStep : Executing step: [generarResumenAnualStep]
2026-08-22T10:18:14.770-04:00 INFO 7596 --- [banco-batch] [ main] c.d.b.listener.BatchStepListener : STEP INICIADO | step=generarResumenAnualStep | fechaInicio=2026-08-22T10:18:14.770674800
2026-08-22T10:18:14.824-04:00 INFO 7596 --- [banco-batch] [ main] o.s.batch.core.step.AbstractStep : Step: [generarResumenAnualStep] executed in 61ms
2026-08-22T10:18:14.837-04:00 INFO 7596 --- [banco-batch] [ main] c.d.b.listener.BatchStepListener : STEP FINALIZADO | step=generarResumenAnualStep | status=COMPLETED | leidos=7 | escritos=7 | omitid
o=1
2026-08-22T10:18:14.883-04:00 INFO 7596 --- [banco-batch] [ main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: {name=estadosAnualesJob}] completed with the following parameters: [[JobParameter{
name='fechaEjecucion', value=20260822101803, type=class java.lang.String, identifying=true}]] and the following status: [COMPLETED] in 317ms
|
```

La captura muestra las métricas de ejecución del procesamiento multihilo utilizado para generar los estados de cuenta anuales.

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXPI_52_Grupo_1\banco-batch> Get-Content ".\output\estados_cuenta_anuales.csv"
cuenta_id,total_depositos,total_retiros,total_compras,saldo_neto,cantidad_movimientos
101,1000.00,-500.00,0,500.00,2
102,1500.00,0,0,1500.00,1
103,2000.00,0,0,2000.00,1
104,0,0,-100.00,-100.00,1
105,2500.00,0,0,2500.00,1
106,3000.00,0,0,3000.00,1
108,2000.00,0,0,2000.00,1
```

La evidencia confirma la creación del archivo CSV con los estados de cuenta anuales generados por el proceso batch.

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
572fbb281e49	postgres:16	"docker-entrypoint.s..."	12 hours ago	Up 26 minutes	0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp	banco-postgres

La captura verifica que el contenedor de PostgreSQL se encuentra activo y disponible para la persistencia de los resultados del sistema.

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> docker exec -it banco-postgres psql -U postgres -d banco_batch -c "SELECT * FROM estados_cuenta_anuales ORDER BY cuenta_id;"
```

cuenta_id	total_depositos	total_retiros	total_compras	saldo_neto	cantidad_movimientos
101	1000.00	-500.00	0.00	500.00	2
102	1500.00	0.00	0.00	1500.00	1
103	2000.00	0.00	0.00	2000.00	1
104	0.00	0.00	-100.00	-100.00	1
105	2500.00	0.00	0.00	2500.00	1
106	3000.00	0.00	0.00	3000.00	1
108	2000.00	0.00	0.00	2000.00	1

(7 rows)

La consulta permite verificar los registros de estados de cuenta anuales almacenados correctamente en PostgreSQL.

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> docker exec -it banco-postgres psql -U postgres -d banco_batch -c "SELECT * FROM intereses_calculados ORDER BY cuenta_id;"
```

cuenta_id	nombre	saldo_inicial	edad	tipo	tasa	interes	saldo_final
101	John Doe	5000.00	30	ahorro	0.0100	50.00	5050.00
102	Jane Smith	8000.00	25	prestamo	0.0200	160.00	8160.00
103	Bob Johnson	12000.00	30	prestamo	0.0200	240.00	12240.00
106	John Doe	5000.00	30	ahorro	0.0100	50.00	5050.00
107	Diana Prince	15000.00	40	prestamo	0.0200	300.00	15300.00
108	Steve Rogers	10000.00	80	ahorro	0.0100	100.00	10100.00

(6 rows)

La evidencia muestra los resultados del cálculo de intereses persistidos en la base de datos PostgreSQL.

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> docker exec -it banco-postgres psql -U postgres -d banco_batch -c "SELECT * FROM transacciones_procesadas ORDER BY id;"
```

id	fecha	monto	tipo
1	2024-01-01	1000.00	debito
2	2024-01-02	1500.00	credito
5	2024-01-04	800.00	credito
6	2024-01-05	700.00	debito
7	2024-01-06	1200.00	credito
8	2024-01-05	700.00	debito
9	2024-01-07	3000.00	debito
10	2024-01-08	1000.00	credito

(8 rows)

La consulta confirma la persistencia de las transacciones válidas procesadas por el Job de Spring Batch.

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> docker exec -it banco-postgres psql -U postgres -d banco_batch -c "SELECT * FROM resumen_transacciones_diarias ORDER BY fecha;"
```

fecha	cantidad_transacciones	total_debitos	total_creditos	monto_total
2024-01-01	1	1000.00	0.00	1000.00
2024-01-02	1	0.00	1500.00	1500.00
2024-01-04	1	0.00	800.00	800.00
2024-01-05	2	1400.00	0.00	1400.00
2024-01-06	1	0.00	1200.00	1200.00
2024-01-07	1	3000.00	0.00	3000.00
2024-01-08	1	0.00	1000.00	1000.00

(7 rows)

La captura muestra el resumen de transacciones almacenado en PostgreSQL, con los totales consolidados por el proceso batch.

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> docker exec -it banco-postgres psql -U postgres -d banco_batch -c "SELECT step_execution_id, step_name, status, read_count, write_count, process_skip_count, commit_count, rollback_count, process_skip_count, commit_count, rollback_count FROM batch_step_execution ORDER BY step_execution_id DESC LIMIT 10;"
```

step_execution_id	step_name	status	read_count	write_count	process_skip_count	commit_count	rollback_count
12	generarResumenAnualStep	COMPLETED	7	7	0	2	0
11	procesarMovimientosAnualesStep	COMPLETED	9	8	1	2	0
10	procesarInteresesStep	COMPLETED	8	6	2	2	0
9	generarResumenTransaccionesStep	COMPLETED	7	7	0	2	0
8	procesarTransaccionesStep	COMPLETED	10	8	2	2	0
7	generarResumenAnualStep	COMPLETED	7	7	0	2	0
6	procesarMovimientosAnualesStep	COMPLETED	9	8	1	2	0
5	procesarInteresesStep	COMPLETED	8	6	2	2	0
4	generarResumenTransaccionesStep	COMPLETED	7	7	0	2	0
3	procesarTransaccionesStep	COMPLETED	10	8	2	2	0

(10 rows)

La consulta a BATCH_STEP_EXECUTION evidencia la ejecución de los Steps y sus estados finales dentro del repositorio de metadatos de Spring Batch.

```

PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> .\mvnw.cmd clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< cl.duoc:banco-batch >-----
[INFO] Building banco-batch 0.0.1-SNAPSHOT
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ banco-batch ---
[INFO] Deleting C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ banco-batch ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 4 resources from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.14.1:compile (default-compile) @ banco-batch ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 18 source files with javac [debug parameters release 21] to target\classes
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time:  4.244 s
[INFO] Finished at: 2026-08-22T10:29:55-04:00
[INFO] -----

```

La evidencia confirma que la construcción del proyecto finaliza correctamente con el resultado BUILD SUCCESS.

```

[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.575 s -- in cl.duoc.bancobatch.BancoBatchApplicationTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time:  8.506 s
[INFO] Finished at: 2026-08-22T10:34:27-04:00
[INFO] -----
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> 

```

La captura acredita la ejecución satisfactoria de la suite de pruebas, sin fallos ni errores reportados.